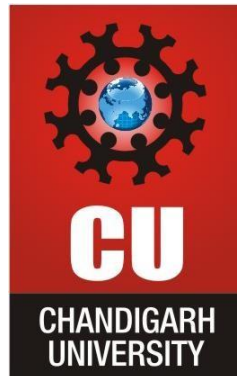




CHANDIGARH UNIVERSITY

Discover. Learn. Empower.



Subject: 24CSH-337

Subject code: Computer Networks

Assignment No.: 1

Student Name: Husanpreet Kaur

Submitted to: Mr. Alok Kumar

UID: 24BCY70138

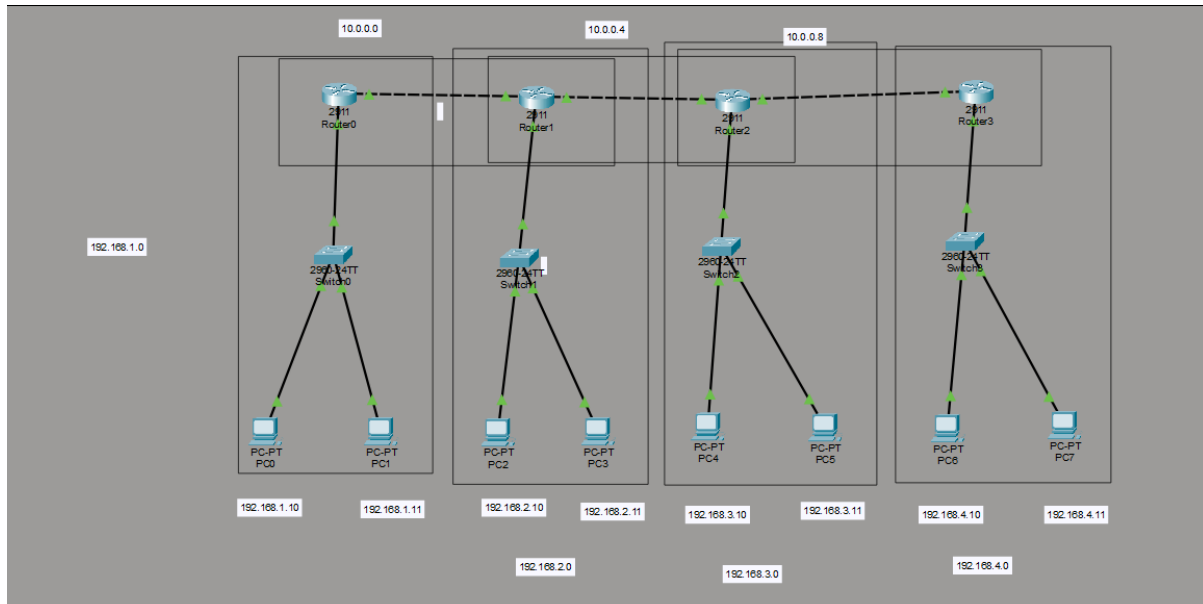
Date of Submission: 09/08/2026

Branch: AIT-CSE(Cyber Security)

Section/Group: 24AIT-KRG2

Q1: Develop Static Routing with 4 PC's, 4 Switches & 8 PC's. Make sure to highlight the Networks with their network ID's. IP Addresses should be written in text format as well.

Sol:



Interface: FastEthernet0

IP Configuration

☒ DHCP ☒ Static

IPv4 Address: 192.168.3.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.3.1

DNS Server: 0.0.0.0

IPv6 Configuration

☒ Automatic ☒ Static

IPv6 Address:

Link Local Address: FE80::290:2BFF:FE9B:D03E

Default Gateway:

DNS Server:

802.1X

☒ Use 802.1X Security

Authentication: MD5

Username:

Password:

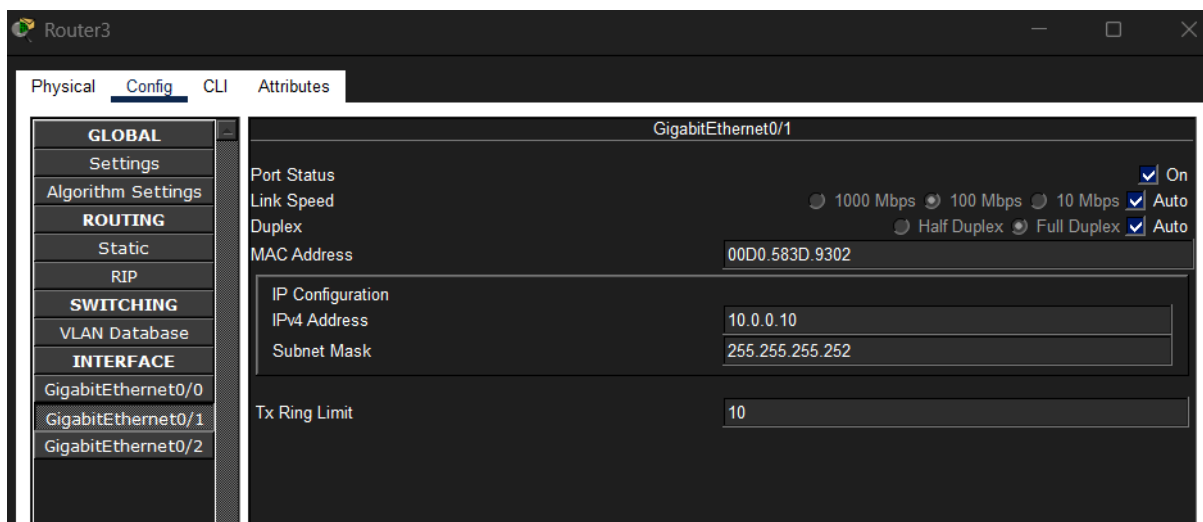
Realtime Simulation										
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	PC3	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC0	PC2	ICMP		0.000	N	1	(edit)	(delete)

Press RETURN to get started!

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 10.0.0.6 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/2
Router(config-if)#ip address 10.0.0.9 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       192.168.3.1    YES manual up          up
GigabitEthernet0/1       10.0.0.6       YES manual up          up
GigabitEthernet0/2       10.0.0.9       YES manual up          up
Vlan1                    unassigned     YES unset  administratively down down
Router#
```



Device	IP Address	Subnet Mask	Default Gateway
Router0 G0/0	192.168.1.1	255.255.255.0	—
PC0	192.168.1.10	255.255.255.0	192.168.1.1
PC1	192.168.1.11	255.255.255.0	192.168.1.1

Device	Interface	IP Address	Subnet Mask
--------	-----------	------------	-------------

Router0	G0/1	10.0.0.1	255.255.255.252
Router1	G0/1	10.0.0.2	255.255.255.252

Device	IP Address	Subnet Mask	Default Gateway
Router1 G0/0	192.168.2.1	255.255.255.0	—
PC2	192.168.2.10	255.255.255.0	192.168.2.1
PC3	192.168.2.11	255.255.255.0	192.168.2.1

Device	Interface	IP Address	Subnet Mask
Router1	G0/2	10.0.0.5	255.255.255.252
Router2	G0/1	10.0.0.6	255.255.255.252

Device	IP Address	Subnet Mask	Default Gateway
Router2 G0/0	192.168.3.1	255.255.255.0	—
PC4	192.168.3.10	255.255.255.0	192.168.3.1
PC5	192.168.3.11	255.255.255.0	192.168.3.1

Device	Interface	IP Address	Subnet Mask
Router2	G0/2	10.0.0.9	255.255.255.252
Router3	G0/1	10.0.0.10	255.255.255.252

Device	IP Address	Subnet Mask	Default Gateway
Router3 G0/0	192.168.4.1	255.255.255.0	—
PC6	192.168.4.10	255.255.255.0	192.168.4.1
PC7	192.168.4.11	255.255.255.0	192.168.4.1

Q2: Write the difference between Hub, Switch & Router in Tabular Format. Explain why router is the most intelligent & Hub is least intelligent device.

Sol:

Basis	Hub	Switch	Router
-------	-----	--------	--------

Operating Layer	Layer 1 – Physical Layer	Layer 2 – Data Link Layer	Layer 3 – Network Layer
Basic Function	Repeats/broadcasts the incoming data to all other ports	Selectively forwards data to the appropriate port	Connects different networks and forwards packets between them
Address Used	Does not read addresses	Uses MAC addresses	Uses IP addresses
Address Table	No address table	Maintains a MAC/CAM address table	Maintains a routing table
Data Forwarding	Blind broadcast	Intelligent/selective forwarding	Path-based forwarding
Data Unit	Bits	Frames	Packets
Collision Domain	All ports share one collision domain	Each port is a separate collision domain	Each interface is a separate collision domain
Broadcast Domain	Single broadcast domain	Normally a single broadcast domain	Each interface separates broadcast domains
Transmission Mode	Half-duplex	Full-duplex	Full-duplex
Learning Ability	Cannot learn device addresses	Learns source MAC addresses	Learns/maintains routes to networks
Path Selection	No	No network-layer path selection	Yes
Network Connectivity	Connects devices in a basic LAN	Connects devices within a LAN	Connects different networks
Traffic Efficiency	Low	High	High
Intelligence	Least intelligent	More intelligent	Most intelligent
Main Advantage	Simple and inexpensive	Efficient LAN communication	Enables communication between different networks
Main Limitation	Sends traffic to all ports and causes collisions	Primarily operates within Layer-2 networks	More complex and requires configuration

(I) Why is a Router the Most Intelligent?

A router is considered the most intelligent among the three devices because it operates at Layer 3 – Network Layer and makes decisions based on logical IP addresses and routing information. The Network Layer is responsible for logical addressing, routing, path selection and packet forwarding.

1. It uses IP addresses

A router examines the **destination IP address** of a packet.

Eg: Network A Network B
192.168.1.0/24 192.168.2.0/24 ——— Router ———

2. It maintains a routing table

A router maintains a **routing table** containing information about destination networks and their paths.

Eg: Destination Network	Next Hop
192.168.2.0/24	10.0.0.2
192.168.3.0/24	10.0.0.6
192.168.4.0/24	10.0.0.10

3. It connects different networks

A hub and switch primarily connect devices within a local network, whereas a router provides **internetworking**, allowing communication between different networks.

4. It performs path selection

A router can determine an appropriate path for packets based on its routing information. Network-layer path selection and forwarding are core functions of a router.

5. It separates broadcast domains

A router acts as a logical boundary between networks. Broadcast traffic from one network is normally not forwarded into another network.

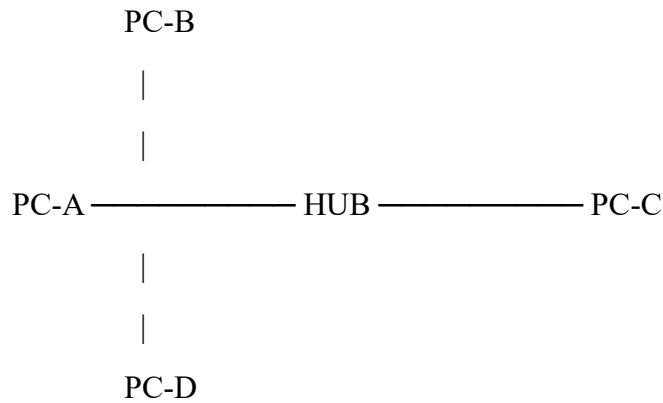
(II) Why is a Hub the Least Intelligent?

A hub is considered the **least intelligent** because it operates only at **Layer 1 – Physical Layer**.

It does not examine MAC or IP addresses and therefore cannot make an intelligent forwarding decision.

Working of a Hub

Suppose PC-A wants to send data to PC-B:



The hub **blindly broadcasts/repeats the signal** to the other active ports.

PC-B accepts the data because it is the intended destination, while PC-C and PC-D discard it because the destination does not match them.

Why is this inefficient?

- It cannot read destination addresses.
- It does not maintain an address table.
- It cannot selectively forward traffic.
- All ports share the same collision domain.
- It normally operates in half-duplex mode.
- Simultaneous transmissions can result in collisions.

(III) Why is a Switch More Intelligent Than a Hub?

A switch operates at **Layer 2 – Data Link Layer** and uses **MAC addresses**.

It maintains a **MAC/CAM address table** that maps MAC addresses to physical ports.

Eg: MAC Address-----Port

AA:AA:AA:AA:AA:AA-----Port 1

BB:BB:BB:BB:BB:BB-----Port 2

CC:CC:CC:CC:CC:CC-----Port 3

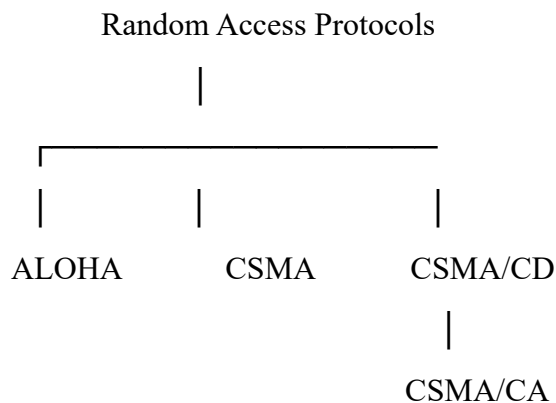
If the switch knows that PC-B's MAC address is on Port 3, it forwards the frame specifically to Port 3. Instead of broadcasting the frame everywhere, it can perform **selective forwarding**.

Q3: What are Random Access Protocols? Why do we need them? Explain each category in detail with diagram and example. Also, explain difference between CSMA/CD & CSMA/CA.

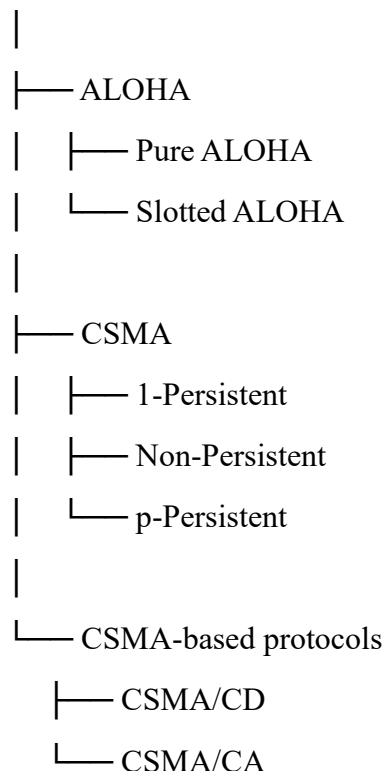
Sol: (i) **Random Access Protocols** are **MAC (Medium Access Control) protocols** used in a shared/broadcast communication channel where multiple devices compete for access to the same transmission medium.

In a random access protocol, **no device has a fixed schedule or predetermined permission to transmit**. A station can attempt to transmit whenever it has data to send, but this may result in a **collision** when two or more stations transmit simultaneously.

The main categories of random access protocols are:



Random Access(In more classified version):



Random access protocols are required when **multiple devices share a common communication medium**. All devices use the same channel. If PC1 and PC3 transmit at exactly the same time, their signals can interfere.

Therefore, random access protocols provide rules for:

- When a device can transmit.
- How a device detects a busy or idle channel.
- How collisions are handled.
- How a device waits before retransmitting.
- How multiple devices share the same medium efficiently.

Main objective:

To allow multiple devices to efficiently and fairly share a common communication channel while reducing collisions and improving channel utilization.

Categories of Random Access Protocols:

A. ALOHA

ALOHA is one of the simplest random access protocols.

The basic idea is:

A station transmits whenever it has data.

It does **not initially check whether the channel is free.**

A1. Pure ALOHA

In **Pure ALOHA**, a station can transmit **at any time.**

Station A ————— DATA —————>
Channel

Station B ————— DATA —————>
↓
COLLISION

If two frames overlap:

Time —————>

A: |----- Frame -----|
(Collision)

B: |----- Frame -----|

The sender waits for an acknowledgement.

If an acknowledgement is not received, the station assumes that a collision occurred and retransmits after a random backoff time.

Advantages

- Very simple.
- No requirement to sense the channel before transmission.
- Suitable for environments where centralized control is difficult.

Disadvantages

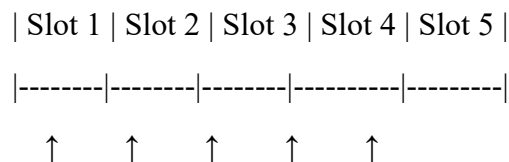
- High probability of collisions.
- Low efficiency.
- Large vulnerable period.
- More retransmissions are required.

A2. Slotted ALOHA

Slotted ALOHA improves Pure ALOHA by dividing time into equal-sized slots.

A station can transmit **only at the beginning of a time slot**.

Time Slots:



Transmission can start only here

Slot 1 | Slot 2 | Slot 3

PC1 —————>

PC2 —————>

↓

Collision

Both detect the unsuccessful transmission and select random future slots for retransmission.

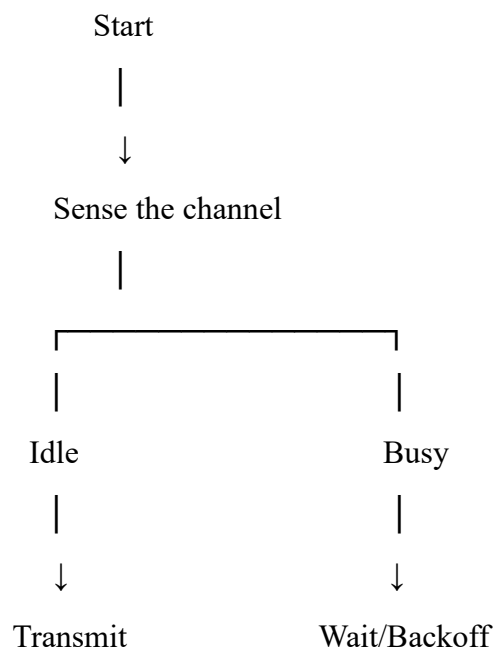
Pure ALOHA	Slotted ALOHA
Transmission can start anytime	Transmission starts only at slot boundaries

Greater collision probability	Lower collision probability
No synchronization required	Requires time synchronization
Less efficient	More efficient

(ii) CSMA – Carrier Sense Multiple Access

CSMA improves upon ALOHA by using **carrier sensing**.

Before transmitting, a station first **checks whether the channel is busy or idle**.



If PC1 is transmitting:

PC1 ————— DATA —————>

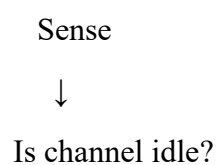
PC2 → Sense channel → BUSY → Wait

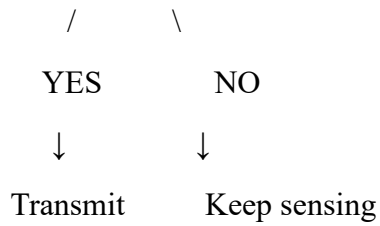
After the channel becomes idle, PC2 can attempt transmission.

Types of CSMA:

1. 1-Persistent CSMA

The station continuously senses the channel.





If the channel is idle:

Transmit immediately.

If the channel is busy:

Keep sensing until it becomes idle.

Advantage

- Simple.
- Low waiting time when the channel becomes free.

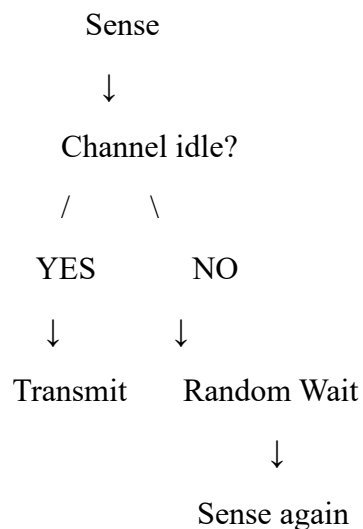
Disadvantage

If several stations are waiting, they may all transmit when the channel becomes idle, causing a collision.

2. Non-Persistent CSMA

If the channel is busy, the station **does not continuously sense it**.

Instead, it waits for a random amount of time and then checks again.



Advantage

Reduces the probability of collisions between waiting stations.

Disadvantage

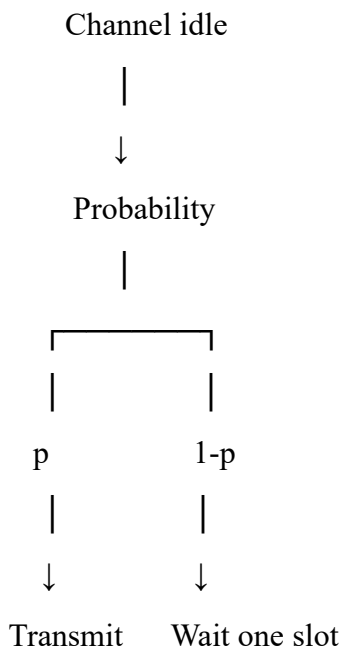
Can introduce greater delay.

3. p-Persistent CSMA

p-Persistent CSMA is generally used with **slotted channels**.

When the channel becomes idle:

- With probability p , the station transmits.
- With probability $1 - p$, it waits for the next slot.



(iii) CSMA/CD:

CSMA/CD stands for:

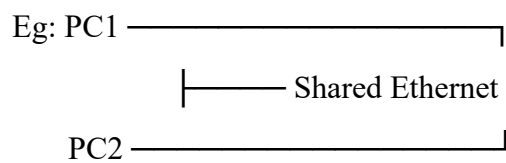
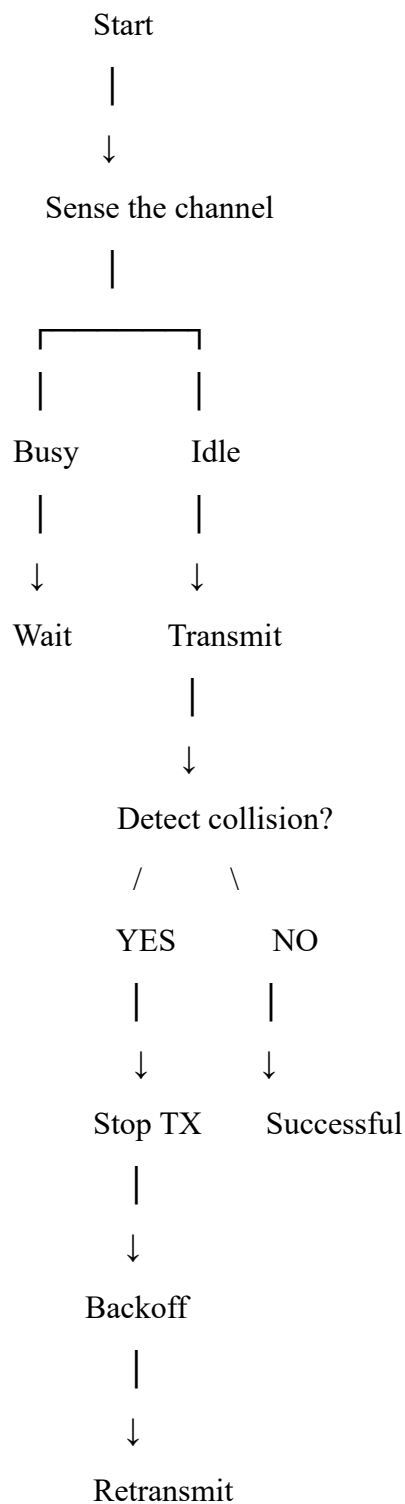
Carrier Sense Multiple Access with Collision Detection

It was traditionally associated with **shared, half-duplex Ethernet**.

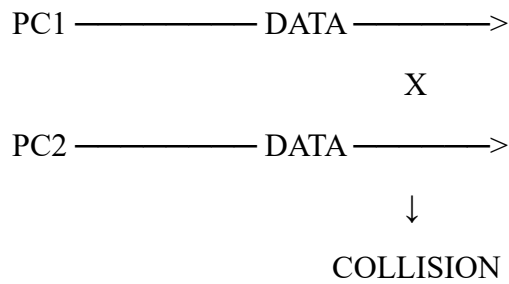
The basic idea is:

1. Sense the channel.
2. If idle, transmit.
3. Continue monitoring while transmitting.
4. If a collision is detected, stop transmitting.
5. Send a jam signal.
6. Wait for a random backoff period.

7. Retransmit.



If both PC1 and PC2 transmit:



Both stop and wait for a random backoff period before attempting again.

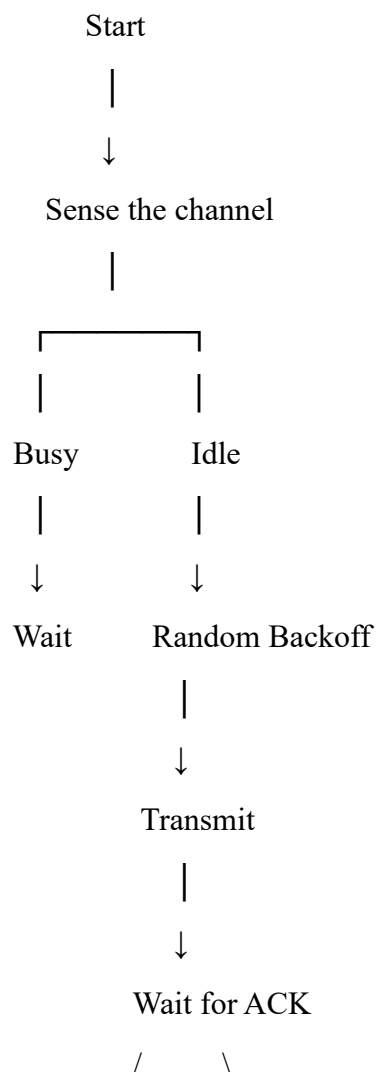
(iv) CSMA/CA:

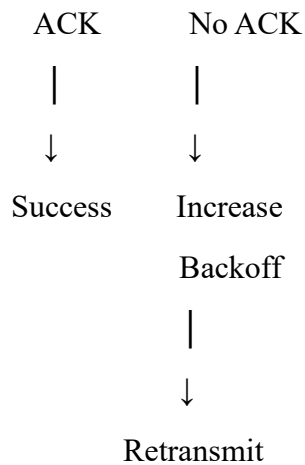
CSMA/CA stands for:

Carrier Sense Multiple Access with Collision Avoidance

It is primarily associated with **wireless LANs such as IEEE 802.11 Wi-Fi**.

Unlike CSMA/CD, a wireless station generally cannot reliably detect a collision while it is transmitting. Therefore, it attempts to **avoid collisions before they happen**.





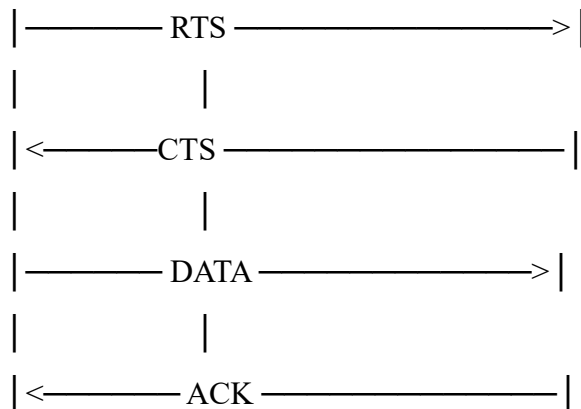
RTS/CTS

Wireless networks may also use:

- **RTS – Request to Send**
- **CTS – Clear to Send**

Eg: Sender

Receiver



RTS/CTS can help reduce collisions caused by the **hidden-terminal problem**.

Basis	CSMA/CD	CSMA/CA
Full Form	Carrier Sense Multiple Access with Collision Detection	Carrier Sense Multiple Access with Collision Avoidance
Main Approach	Detects collisions	Attempts to avoid collisions
Commonly Associated With	Traditional shared Ethernet	Wireless LAN / Wi-Fi

Collision Handling	Detects collision during transmission	Uses mechanisms such as backoff and acknowledgements to reduce collision probability
Transmission	Transmits when channel is sensed idle	Senses channel and uses contention/backoff before transmission
ACK	Not normally required for basic Ethernet collision detection	ACK is commonly used in Wi-Fi
Collision Detection	Possible in traditional shared Ethernet	Difficult/impractical for a wireless station while transmitting
Backoff	Used after collision	Used before/repeatedly after unsuccessful transmission
RTS/CTS	Not normally used	Can be used
Main Goal	Detect and recover from collisions	Avoid collisions as much as possible
Medium	Shared wired Ethernet	Wireless medium
Example	Older hub-based Ethernet	IEEE 802.11 Wi-Fi

Why can't Wi-Fi normally use CSMA/CD?

In wired Ethernet, a device can monitor the medium while transmitting and detect a collision.

In wireless communication, a station's own transmitted signal is much stronger at its receiver than another station's signal. Therefore, reliably detecting a collision while transmitting is difficult.

So instead of: Transmit → Detect Collision → Stop

Wi-Fi generally uses: Sense → Wait/Backoff → Transmit → ACK

Therefore:

Ethernet traditionally uses CSMA/CD for shared half-duplex communication, while Wi-Fi uses CSMA/CA to reduce the probability of collisions.

Modern switched full-duplex Ethernet normally does **not require CSMA/CD**, because each link is dedicated and collisions do not occur in normal full-duplex operation.